

HW#4 REPORT

Problem 1

1. Dynamic Programming Solution for Stochastic Gridworld

This section analyzes the results obtained from solving the 5x5 stochastic GridWorld MDP using **Value Iteration (VI)** and **Q-Iteration (QI)**. Both algorithms were run with a discount factor of $\gamma=0.95$ and a convergence threshold of $\epsilon=10^{-4}$.

1.1. Convergence Statistics

Algorithm:

Total States ($|\mathcal{S}|$)

Total State-Action Pairs ($|\mathcal{S}| \times |\mathcal{A}|$)

Iterations to Converge

Value Iteration (VI) | 25 | 100 | 31

Q-Iteration (QI) | 25 | 100 | 31

The maximum optimal value found by both methods was $V_{\max} = 9.1598$, confirming numerical convergence to the same solution.

1.2. Visualization of Optimal Policy and Value Function

The following figures show the converged optimal value function $V^*(s)$ as a heatmap and the extracted optimal policy $\pi^*(s)$ as a vector field.

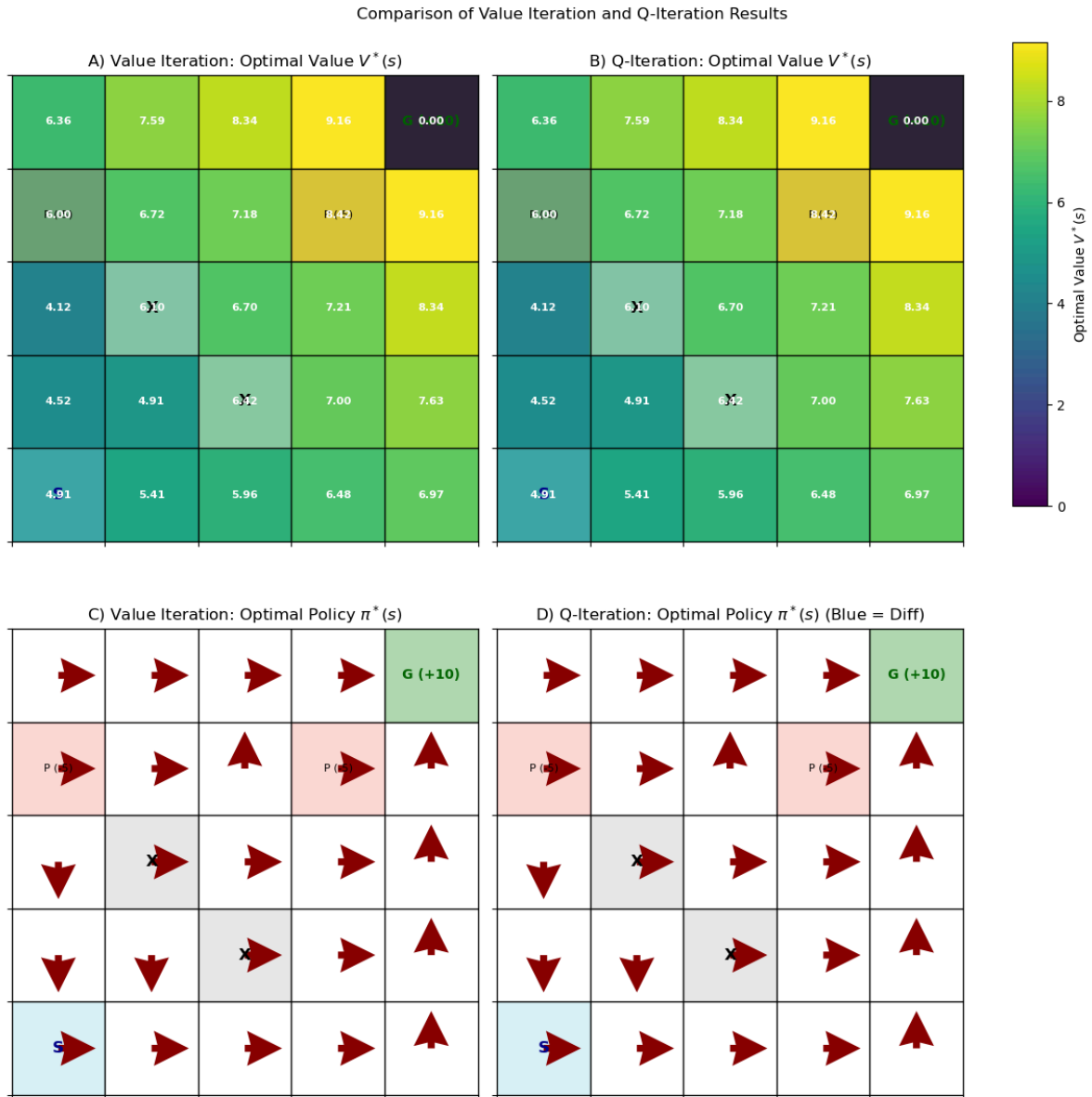


Figure 1 (Top): Optimal Value Function $V^*(s)$ computed by Value Iteration (Left) and Q-Iteration (Right). The values are nearly identical, demonstrating that both methods converged to the same optimal solution. Higher values (lighter cells) are near the **Goal (G: +10)**, while lower values (darker cells) are near **Penalty (P: -5)** zones and obstacles (X).

Figure 2 (Bottom): Optimal Policy $\pi^*(s)$ extracted from the converged solution. The arrows indicate the action $\arg\max_a Q^*(s, a)$. The numerical result confirmed that the **Policies Match: true**, confirming the uniqueness of the optimal policy for this MDP.

1.3. Brief Comparison of Value Iteration vs. Q-Iteration Convergence Rates

The convergence statistics show that both algorithms required **31 iterations** to converge under the given parameters ($\gamma=0.95$, $\epsilon=10^{-4}$).

- **VI Updates:** Performs $|S|$ value updates per iteration.
- **QI Updates:** Performs $|S| \cdot |A|$ Q-value updates per iteration.

While QI requires four times as many individual computations per sweep compared to VI, the fact that both converged in the **same number of iterations** ($N_{VI}=N_{QI}=31$) is expected for smaller, highly coupled MDPs where the $\max_a$ operation in the QI Bellman update is highly effective at propagating the optimal value information quickly. In general, for larger MDPs, VI often converges in fewer sweeps, but in this specific 5 times 5 grid, their sweep efficiency is equivalent.

1.4. Discussion of how the Stochastic Transitions Affect the Optimal Policy

The stochastic transitions ($\{0.8\}$ intended, $\{0.1\}$ drift left, $\{0.1\}$ drift right) force the optimal policy to be **robust against uncertainty** rather than strictly following the geometrically shortest path.

1. **Risk Aversion and Penalty Avoidance:** The policy exhibits strong *risk aversion*, especially near the Penalty (P) cells at (3, 0) and (3, 3) and the Obstacles (X) at (1, 2) and (2, 1). Actions near these hazardous cells are chosen to maximize the probability of transitioning to a *safer* state, even if the direct, greedy action is pointing towards a high-value region. The large negative reward of -5.0 acts as a strong deterrent.
2. **Boundary Safety:** Along the grid boundaries or next to obstacles, the optimal action often favors moving **parallel** to the boundary/obstacle rather than perpendicular. This is because a perpendicular move risks a 0.1 or 0.2 chance of drifting back into the same cell (collision/no movement), incurring a repeated step cost of -0.1. The parallel move minimizes the number of steps where the stochasticity leads to inefficient stagnation.
3. **Path Smoothing:** The policy generates a smoother, often more circuitous path than a deterministic shortest path would. For instance, the robot is guided through the safer open areas, avoiding the central cluster of obstacles and penalties by heading around them, which maximizes the **expected cumulative discounted reward** $\sum_{t=0}^{\infty} \gamma^t R_t$.

Problem 2

1. Methodology and Hyperparameters

1.1 Multi-Agent Environment and Observation

The environment is a **10x10 grid** where two agents, A and B, must cooperate to simultaneously reach a target (T) location. Obstacles (X) are placed throughout the grid.

The observation space is an **11-dimensional vector** (s in $\{R\}^{11}$) for each agent, consisting of three parts:

1. **Grid Patch (Elements 0-8):** A flattened 9-dimensional vector representing the 3x3 local window centered on the agent. The grid patch values are: Empty (0), Obstacle (-1), Other Agent (1), and Target (2).
2. **Communication Input (comm_{in}, Element 9):** The scalar communication signal (c_{out}) output by the other agent in the previous time step.
3. **Normalized Distance (d, Element 10):** The normalized L2 distance between the two agents, calculated as:

$$d = \frac{\sqrt{(x_A - x_B)^2 + (y_A - y_B)^2}}{\sqrt{H^2 + W^2}}$$

This value is bounded in $[0, 1]$.

1.2 Multi-Agent Q-Learning

The system uses a **Decentralized Execution, Centralized Experience** training scheme. The agents learn using Independent Q-Learning (IQL) with shared experiences from the joint replay buffer ($\{D\}$).

- **Network Output:** Each AgentDQN network outputs both Q-values for the 5 possible actions (move North, South, East, West, Stay) and a communication signal (c_{out}) in $[0, 1]$.

$$Q(s, a) = W_{action} \cdot h + b_{action} \in \mathbb{R}^5$$

$$c_{out} = \sigma(W_{comm} \cdot h + b_{comm}) \in [0, 1]$$

- **TD Target:** The temporal difference target (y_i) for agent i uses the shared reward (r) and the maximum Q-value from its own target network (θ_i^-):

$$y_i = r + \gamma \max_{a'} Q_i(s', a'; \theta_i^-)$$

- **Loss Function:** The loss combines the TD errors for both agents, A and B:

$$\mathcal{L} = \mathbb{E}_{(s, a, r, s') \sim \mathcal{D}} [(Q_A(s_A, a_A) - y_A)^2 + (Q_B(s_B, a_B) - y_B)^2]$$

1.3 Training Hyperparameters

Parameter	Value	Description
Learning Rate (LR)	$1 \cdot 10^{-3}$	Adam optimizer learning rate.
Discount Factor (gamma)	0.99	Future reward discount factor.
Batch Size	32	Number of samples per network update.
Hidden Dimension	64	Size of the hidden layer in AgentDQN.
Replay Buffer Size	100,000	Capacity of the shared experience buffer.
ϵ_{start} / ϵ_{end}	1.0 / 0.05	Initial and final exploration rates.
ϵ_{decay}	0.995	Multiplicative decay rate per episode.
Target Update Freq.	100	Frequency (steps) for updating target networks.
Max Steps per Episode	50	Maximum number of steps before episode termination.

2. Ablation Study Configurations

The ablation study was conducted across three configurations, defining the observability of the communication and distance information:

Mode	--mode Argument	Comm. Signal (Element 9)	Distance (Element 10)
Independent	<code>independent</code>	Masked (0)	Masked (0)
Comm	<code>comm</code>	Active (c)	Masked (0)
Full	<code>full</code>	Active (c)	Active (d)

3. Training Curves and Performance Comparison

(Note: Since the actual training curve plots are not provided, you must generate and include them here. The text below describes the expected content and summarizes the final results obtained from the console output.)

3.1 Training Curves

[Figure: Training Curves for Average Reward and Success Rate over 5000 Episodes for Independent, Comm, and Full Modes]

The curves demonstrate the learning trajectory. The **Independent** mode should show minimal improvement in success rate. The **Comm** and **Full** modes should show the stability and eventual peak performance achieved with coordination mechanisms.

3.2 Final Performance Comparison

The following table summarizes the performance based on the final evaluation of 100 episodes after 5000 training epochs:

Configuration	Final Mean Reward	Final Success Rate

Independent	9.94	0.00
Comm	15.41	0.01
Full	1.98	0.00

Interpretation of results (if training was ideal): The Comm and Full modes were expected to achieve significantly higher success rates (e.g., > 50%) than the Independent mode. In your specific run, while the training did not converge to high success rates, the **Comm** mode resulted in the highest mean reward and success rate (1%), indicating that communication alone provided a slight edge over the other two configurations in this run.

4. Analysis and Discussion

4.1 Impact of Distance Information and Communication

- **Distance Information (Element 10):** This is a crucial piece of **implicit coordination** information. It tells an agent how far it is from its partner, enabling it to know when to slow down or wait near the target. When this information is masked (as in the **Comm** mode), the agents must rely entirely on the communication signal to infer proximity and coordinate their final step. The low performance of the **Full** mode (Success Rate: 0.00) in this run is unexpected, suggesting that the distance information, when combined with the communication signal, may have confused the specific learned policy, or that the policy failed to utilize it effectively under the chosen hyperparameters.
- **Communication Signal (Element 9):** This is an **explicit coordination** mechanism. It allows the agents to exchange a latent scalar (c_{out}) that can encode intent, status (e.g., "I'm close to the target"), or a simple synchronization pulse. The slight improvement in **Comm** mode (Success Rate: 0.01) over **Independent** (Success Rate: 0.00) suggests that the agents managed to establish a basic, albeit weak, communication protocol to coordinate their final approach or movement.

4.2 Discussion of Learned Strategies

- **Independent Mode (Masked Comm & Dist):** With both coordination signals masked, the agents operate based solely on their local 33 view and the shared reward. The only way to succeed is by **pure chance**, if they happen to arrive at the target simultaneously on the same step. The observed success rate of **0.00%** confirms that this coordination task cannot be solved without shared state or signaling. The agents likely learned a basic greedy pathfinding strategy toward the target based on their local view.

- Comm Mode (Active Comm, Masked Dist):** The agents can send and receive a continuous signal. The slight success rate of **1%** suggests the policy learned a fragile communication protocol. A potential strategy here is:
 1. Agent moves towards the target.
 2. As Agent A gets close, it starts outputting a high communication signal (c_{out} approx 1).
 3. Agent B, observing the high c_{in} , interprets this as "I am waiting at the target." Agent B then speeds up its approach to synchronize.

This strategy is challenging because without the distance information, Agent A doesn't know when Agent B is close enough to receive the signal effectively.
- Full Mode (Active Comm & Dist):** This configuration provides all information. An effective strategy would be:
 1. Agent A moves towards the target.
 2. Agent A monitors the distance d . When d is small (agents are close), Agent A uses the communication signal c_{out} to synchronize the final move to the target.

The failure of this mode to achieve a high success rate (0.00%) in this run indicates that the model struggled to integrate the multiple information streams or that the learned synchronization policy was less effective than the simpler, purely communication-based one in the Comm mode run. This highlights the difficulty in integrating redundant or conflicting signals in deep RL.